Improving the covering code bound $K(1,7,1) \le 48$ requires an asymmetric code

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October 2026

Abstract

Let $K(1,7,1)$ be the smallest number of words in a covering code of radius 1 in the mixed Hamming space $\mathbb{F}_3 \times \mathbb{F}_2^7$. The tables of mixed binary/ternary covering codes give $42 \le K(1,7,1) \le 48$, and the upper bound comes from the direct product of $\mathbb{F}_3$ with the binary Hamming code of length 7. We prove, with the help of a computer, that every covering code of radius 1 in this space with at most 47 words has trivial automorphism group. The proof reduces the question to one representative of each of the 46 conjugacy classes of isometries of prime order and solves an integer program for each; the minimum size of an invariant covering code turns out to be 48, 54 or 56 in every case. Two independent integer-programming solvers agree on all 46 cases.

1 Introduction

A *covering code* of radius R in a Hamming space is a set of words such that every word of the space is within Hamming distance R of at least one of them. For mixed spaces $\mathbb{F}_3^t \times \mathbb{F}_2^b$ the smallest possible size is denoted $K(t,b,R)$; bounds are collected in tables maintained by Kéri [4], following [6, 1], and the general theory is in the monograph [2].

This note concerns the entry $K(1,7,1)$, for which the tables give

$$42 \le K(1,7,1) \le 48. \tag{1}$$

The lower bound is due to Lang, Quistorff and Schneider [5]. The upper bound is the elementary construction $K(t,b,R) \le 3K(t-1,b,R)$: if $H \subseteq \mathbb{F}_2^7$ is the Hamming code, which has 16 words and covering radius 1, then $\mathbb{F}_3 \times H$ has 48 words and covering radius 1. This code is highly symmetric.

A standard way to look for good codes is to prescribe a group of symmetries and search only among codes it leaves invariant; Östergård and Weakley used this to construct covering codes [7]. Here the same idea is used in the opposite direction: to show that a code improving the upper bound in (1) cannot have any symmetry at all.

Theorem 1. *Let $C \subseteq \mathbb{F}_3 \times \mathbb{F}_2^7$ be a covering code of radius 1 with $|C| \le 47$. Then the only isometry of $\mathbb{F}_3 \times \mathbb{F}_2^7$ that maps C onto itself is the identity.*

The proof is computer-assisted (Section 4). It also determines, for each type of symmetry of prime order, the smallest covering code having that symmetry (Table 1).

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2 Preliminaries

Write $\mathcal{S} = \mathbb{F}_3 \times \mathbb{F}_2^7$; it has $3 \cdot 2^7 = 384$ words. The Hamming distance $d(x, y)$ is the number of coordinates in which x and y differ, and $B(x) = \{y \in \mathcal{S} : d(x, y) \leq 1\}$. Each ball has $1 + 2 + 7 = 10$ words.

Definition 2. An *isometry* of $\mathcal{S}$ is a bijection $\mathcal{S} \to \mathcal{S}$ that preserves d. The isometries form a group G. For a code $C \subseteq \mathcal{S}$, its *automorphism group* is $\mathrm{Aut}(C) = \{g \in G : g(C) = C\}$.

The isometries of a Hamming space are exactly the maps that permute coordinates carrying alphabets of the same size and apply a permutation of the alphabet in each coordinate [2]. Here the ternary coordinate cannot be moved, so

$$G \;\cong\; S_3 \times (S_2 \wr S_7), \qquad |G| = 6 \cdot 2^7 \cdot 7! = 3\,870\,720.$$

An element of $S_2 \wr S_7$ is a pair (f, σ): a permutation σ of the seven binary coordinates followed by flipping the coordinates in a set f.

3 Reduction to finitely many integer programs

Lemma 3. *If C is a covering code of radius 1 and $h \in G$, then $h(C)$ is a covering code of radius 1 with $|h(C)| = |C|$, and $\mathrm{Aut}(h(C)) = h\,\mathrm{Aut}(C)\,h^{-1}$.*

Proof. Since h is a bijection preserving distance, $h(B(c)) = B(h(c))$, so the balls around $h(C)$ are the images of the balls around C and still cover $\mathcal{S}$. Further, $g(C) = C$ if and only if $(hgh^{-1})(h(C)) = h(C)$. □

Lemma 4. *To prove Theorem 1 it suffices to show: for one representative r of each conjugacy class of elements of prime order in G, no covering code C of radius 1 with $|C| \leq 47$ satisfies $r(C) = C$.*

Proof. Suppose $|C| \leq 47$ and $\mathrm{Aut}(C)$ contains some $g_0 \neq \mathrm{id}$. Since G is finite, g_0 has some order $n > 1$. Choose a prime p dividing n; then $g = g_0^{n/p}$ has order exactly p, and $g \in \mathrm{Aut}(C)$ because every power of an automorphism is an automorphism. Let $r = hgh^{-1}$ be the chosen representative of the conjugacy class of g. By Lemma 3, $h(C)$ is a covering code of the same size and $r \in \mathrm{Aut}(h(C))$, contradicting the assumption. □

Lemma 5. *The elements of prime order in G fall into exactly 46 conjugacy classes: 39 of order 2, 5 of order 3, one of order 5 and one of order 7. Representatives are $(\alpha, (f, \sigma))$ where*

- *$p = 2$: α is the identity or a transposition of $\mathbb{F}_3$; σ is a product of $c \in \{0, 1, 2, 3\}$ disjoint transpositions; f flips k of the $7 - 2c$ coordinates fixed by σ, where $0 \leq k \leq 7 - 2c$; and not all of α, c, k are trivial;*
- *$p = 3$: α is the identity or the 3-cycle $x \mapsto x + 1$; σ is a product of $c \in \{0, 1, 2\}$ disjoint 3-cycles; $f = \emptyset$; and not both α and c are trivial;*
- *$p = 5$: $\alpha = \mathrm{id}$, σ a 5-cycle, $f = \emptyset$;*
- *$p = 7$: $\alpha = \mathrm{id}$, σ a 7-cycle, $f = \emptyset$.*

Proof. An element (α, β) of the direct product has prime order p exactly when $\alpha^p = 1$, $\beta^p = 1$ and they are not both the identity, and it is conjugate to (α', β') exactly when α is conjugate to α' in S_3 and β to β' in $S_2 \wr S_7$. In S_3 the classes are the identity, the transpositions and the 3-cycles.

Conjugacy classes in $S_2 \wr S_7$ are determined by the cycle type of σ together with, for each cycle, the parity of the number of flipped coordinates on it [3, Ch. 4]. If $\beta = (f, \sigma)$ has $\beta^p = 1$ then $\sigma^p = 1$, so every cycle of σ has length 1 or p. On a cycle of length p, β^p flips each coordinate of the cycle once for every flip on the cycle, so the parity must be even; such a cycle is conjugate to one with no flips. On a fixed coordinate a flip has order 2, so flipped fixed coordinates occur only when $p = 2$. This gives the representatives listed, and counting them gives $2 \cdot (8 + 6 + 4 + 2) - 1 = 39$ classes of order 2, $3 + 2 = 5$ of order 3, and one each of orders 5 and 7. As an independent check, the program computes the same counts by brute-force enumeration of the whole group on five smaller spaces (`tests/test_classify.py`). □

The integer program. Fix a representative r, and let $O_1, \dots, O_m$ be the orbits of the cyclic group $\langle r \rangle$ on $\mathcal{S}$, with representatives $w_i \in O_i$. A code is r-invariant exactly when it is a union of orbits. If C is r-invariant then a word w is covered if and only if $r(w)$ is, because r is an isometry; so it suffices to check coverage at $w_1, \dots, w_m$. Hence the smallest r-invariant covering code has size

$$m(r) \;=\; \min\Big\{ \sum_{j=1}^{m} |O_j|\, y_j \;:\; \sum_{j \colon O_j \cap B(w_i) \neq \emptyset} y_j \geq 1 \ (1 \leq i \leq m),\ y \in \{0,1\}^m \Big\}. \tag{2}$$

By Lemmas 4 and 5, Theorem 1 is equivalent to $m(r) \geq 48$ for all 46 representatives.

4 Computation

The programs (2) have between 60 and 288 variables. They were solved twice, by separately written code:

1. **First pass.** `src/classify.py` builds each program from the class representative and solves it with HiGHS (through SciPy 1.16.3), either minimising and reading the proven lower bound or adding the constraint $\sum |O_j| y_j \leq 47$ and obtaining infeasibility. All 46 classes were excluded; total solver time 2655 s.

2. **Second pass.** `src/recheck.py`, which shares no code with the first pass, replaces each representative r by a random conjugate hrh^{-1} (seed 2026), rebuilds the orbits and the program from scratch, and minimises (2) with SCIP 10.0 through PySCIPOpt 6.3.0. SCIP proved an optimal value $m(r) \in \{48, 54, 56\}$ in all 46 cases; total solver time 1715 s on one core of an Intel Core Ultra 7 255HX.

Before the second pass, the method was tested on a case with a known answer: for two classes of order 2, allowing 48 words, SCIP produced an invariant covering code of size 48, and an independent brute-force checker (`src/checker.py`) confirms that both codes cover all 384 words (`results/codes/K_1_7_1_M48_scip_*.txt`). Partial runs with two further solvers, CBC on 14 classes and HiGHS on a random conjugate for 11, agree as well. Table 1 lists every class with both results.

Remark 6. Both solvers work in floating-point arithmetic, so their verdicts are strong evidence rather than certificates of the kind an exact verifier would produce. Two solvers written by different teams, applied to independently built instances, agree on every case.

5 Discussion

For many classes $m(r) = 48$: the record 48 can be reached by codes with many different symmetries, but not one of them can be shrunk while keeping its symmetry. Theorem 1 therefore says that a search for a 47-word code can ignore all symmetric codes, which is where searches usually start. Whether $K(1, 7, 1) = 48$ remains open; deciding it requires handling the asymmetric codes, for which the reduction above gives no help.

Code and data. All programs, logs and the generator of Table 1 are at `https://github.com/Noah-Parsons/covering-codes`.

References

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A The 46 classes

Table 1: One row per conjugacy class of prime-order isometries. α: map on the ternary coordinate; c: number of p-cycles on binary coordinates; k: flipped fixed binary coordinates; orbits: number of variables in (2). HiGHS: proven lower bound when the first pass minimised ("–" where it proved the capped program infeasible). SCIP: proven optimum $m(r)$ on a random conjugate. Times in seconds.

#	p	α	c	k	orbits	HiGHS	time	SCIP $m(r)$	time
1	7	id	1	0	60	–	0.0	48	0.1
2	5	id	1	0	96	–	0.0	56	0.4
3	3	$x \mapsto x+1$	0	0	128	–	0.0	48	0.0
4	3	$x \mapsto x+1$	1	0	128	–	0.7	48	1.3
5	3	$x \mapsto x+1$	2	0	128	–	1.2	48	1.6
6	3	id	1	0	192	–	3.7	54	41.9
7	3	id	2	0	144	–	5.6	48	1.8
8	2	(1 2)	0	0	256	48.00	11.0	48	6.1
9	2	(1 2)	0	1	192	48.00	3.1	48	2.3
10	2	(1 2)	0	2	192	48.00	17.6	48	10.4
11	2	(1 2)	0	3	192	48.00	163.7	48	24.9
12	2	(1 2)	0	4	192	48.00	48.4	48	19.0
13	2	(1 2)	0	5	192	48.00	92.6	48	25.1
14	2	(1 2)	0	6	192	48.00	56.2	48	24.4
15	2	(1 2)	0	7	192	48.00	97.0	48	26.1
16	2	(1 2)	1	0	224	48.00	41.4	48	30.0
17	2	(1 2)	1	1	192	48.00	17.5	48	13.1
18	2	(1 2)	1	2	192	48.00	83.0	48	28.7
19	2	(1 2)	1	3	192	48.00	93.2	48	35.4
20	2	(1 2)	1	4	192	48.00	73.5	48	70.0
21	2	(1 2)	1	5	192	48.00	100.7	48	30.9
22	2	(1 2)	2	0	208	48.00	173.7	48	21.9
23	2	(1 2)	2	1	192	48.00	43.0	48	14.2
24	2	(1 2)	2	2	192	48.00	107.3	48	35.8
25	2	(1 2)	2	3	192	48.00	133.5	48	33.8
26	2	(1 2)	3	0	200	48.00	42.5	48	52.5
27	2	(1 2)	3	1	192	48.00	22.6	48	35.8
28	2	id	0	1	192	–	1.7	48	2.0
29	2	id	0	2	192	–	1.7	48	2.2
30	2	id	0	3	192	48.00	37.3	48	23.0
31	2	id	0	4	192	48.00	29.8	48	14.7
32	2	id	0	5	192	48.00	84.8	48	37.0
33	2	id	0	6	192	48.00	53.8	48	26.6
34	2	id	0	7	192	48.00	56.0	48	11.2
35	2	id	1	0	288	48.00	192.3	48	626.4
36	2	id	1	1	192	–	6.2	48	16.0
37	2	id	1	2	192	–	7.6	48	16.4
38	2	id	1	3	192	48.00	62.3	48	52.5
39	2	id	1	4	192	48.00	94.9	48	24.1
40	2	id	1	5	192	48.00	75.7	48	60.5
41	2	id	2	0	240	48.00	176.6	48	70.9
42	2	id	2	1	192	48.00	34.9	48	15.4

#	p	α	c	k	orbits	HiGHS	time	SCIP $m(r)$	time
43	2	id	2	2	192	48.00	25.2	48	14.7
44	2	id	2	3	192	48.00	119.7	48	40.5
45	2	id	3	0	216	48.00	105.2	48	27.7
46	2	id	3	1	192	48.00	57.0	48	45.8